

[DC] The impact of the Informational load of Presence Illusions on Users Attention and Memory.

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ABSTRACT

Presence has become an expected outcome for most VR experiences due to its positive relationship with engagement, task performance, behavior change, and other experiences. However, current literature reports contradictory results regarding presence, cognitive load, working memory, and attention. This study explores the cognitive load of presence under the framework of illusions to understand how these illusions impact users' attention and memory. Quantify and hierarchize the information built by presence (Place and Plausibility of illusion), attempt to contribute knowledge to further design, and effectively manage attention and distraction on Virtual reality experiences. This document discussed our theoretical direction and a proposed psychophysiological study.

Keywords: VR Presence, Attention, Memory, place illusion (PI), plausibility illusion (Psi), immersion, coherence.

Index Terms: human factors, physiology, user experience, virtual reality.

1 INTRODUCTION

Virtual Reality (VR) presence has a variety of definitions, approaches, and theoretical models; however, it is commonly defined as the sensation of being there [1]–[4] or as the sense of feeling inside of a displayed human developed, virtual environment [5].

According to the framework proposed by Slater [6], [7], the construct of presence can be understood as a set of two main illusions, which are place illusion (PI) and plausibility of illusion (PSI).

Current research extending PI and PSI framework [8], [9], establishes an articulation of the concepts of presence, immersion, plausibility, and place illusion. Their work defines that PI is an individual's psychological answer from immersion, and PSI might be the individual's perceptual answer from coherence.

The literature explains that alterations in the VR system impact participants' immersion level, which alters the reported levels of PI. Parallely, modifications on the scenario express changes in the perceived coherence of users; this impacts the capability of users to feel the illusion of the virtual world "plausible" PSI.

The alterations or changes in objective factors such as VR systems or VR scenarios produce changes in the levels of presence and in the user's perception of illusions. Factors of coherence are reported in literature, such as virtual body, non-player character behavior, physical coherence, and scenario characteristics [8]. Also, some of the immersion factors of presence that impact PI are field of view, haptics, spatial sound, frame rate, and synchrony of response among others [7], [9].

2 FIELD PROBLEM

The extension of research on the field shows that the construction of presence is in relationship with conflicting concepts such as

cognitive load, working memory capacity, and long-term memory in a variety of experiences [10], [11].

While some studies [12] establish an inversely proportional relationship between presence and working memory. Other studies [10], [11] demonstrated a directly proportional correlation between presence and cognitive load. The latter has been reported in current findings by Skarbez 2021 [9], but not directly investigated in a specific study.

The current literature could not explain in detail which dimension of the construct of presence (either PI or PSI) is causing overload and in which proportion the factors of PI and PSI are operating. In the meantime, presence factors of PI or PSI have not been completely tested in terms of attention and memory.

We believe that the form in which the illusions are built and presented to users in the stimuli impacts differently their levels of presence, attention, and memory. The information from PI and Psi factors should be responsible for part of the cognitive load reported by the abovementioned studies. The load produced by presence illusions compete with the information of the central task generating cognitive overload. This is happening in two situations. First when the illusion is built lower the levels of expectation of the user. Second, when the illusions are presented all at the same time in the stimuli. We call inconsistencies of the stimuli when users are perceiving as an unusual element or inappropriate according to their expectation.

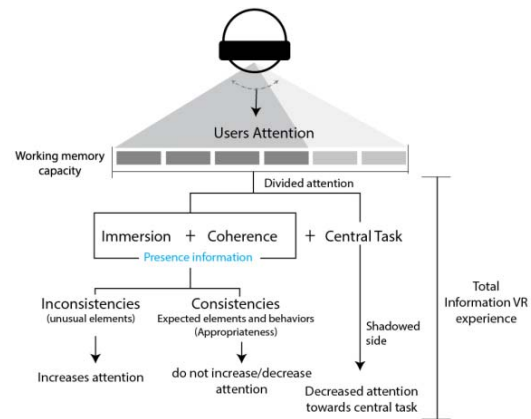


Figure 1: Framework exploration on presence and attention

We also theorize that introducing these inconsistencies simultaneously has a different effect on attention and memory than presented one by one. This would make difference in the overall reported level of presence and in the working memory levels. Where the perceptual adaptation process could operate better and accept the inconsistency as real.

We hypothesize that inconsistencies in the coherence and immersion have a similar impact drawing user attention. However, have a different one on the overall levels of presence [9], [13]. We propose to clarify how the inconsistencies of presence stimuli PI and PSI are impacting attention of users. The paradigm of PI and PSI including the factors defined by Skarbez et al., [6], [8] will aim to identify the cognitive load elements of presence. We Are

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currently extending the illusions framework from the optical of attention, information, and presence illusions.

3 STUDY PROPOSAL

This study proposes to extend the literature on the paradigm of VR presence illusions elucidating how elements of presence, either PSI or PI, and their factors impact users' attention restricting concentration to the central task and using their memory. Two primary studies are proposed to address each dimension of presence, one study for Immersion and PI, and one study for Coherence and PSI.

The study's main objective is to investigate the informational elements of presence illusions that impact users' attention and memory in virtual reality environments. Including three main activities:

To test the effects of PSI and PI on users' attention to elucidate the presence factors that impact users working memory.

Identify and quantify the hierarchy of these elements according to their attentional impact.

Provide guidelines for appropriate informational design according to their cognitive impact.

4 RESEARCH QUESTIONS

After this, we formulated some research questions to address within the study.

Q1. How do the inconsistencies of immersion as VR system and coherence as VR scenario impacts users' attention? They impact similarly or differently?

Q2. How do the coherence factors of presence and PSI impact users' attention on the virtual environment?

Q3. How do the immersion factors of PI impact users' attention on the virtual environment?

Q4. How do the impact of attention of the inconsistencies presented in presence illusions affect the overall levels of presence.

We formulated some hypotheses for the first study. We expect that Inconsistencies produced by immersion and coherence will impact users' attention similarly. However, their impact on presence would be different due to the nature of the effects of place and plausibility of illusion.

H1. Inconsistencies on the coherence stimuli of the virtual scenario will lead to higher levels of attention and higher levels of presence

H2. Inconsistencies on the immersion of the virtual system will lead to higher levels of attention and lower levels of presence

5 AIM OF THE STUDY

This research project clarifies the effects of PI, PSI, coherence and immersion on attention and memory in a virtual environment over human subjects. The above aims to identify how alterations of each factor of presence are drawing users' attention.

Understanding the effects of these factors contributes to designing appropriate content on VR, avoiding excessive informational and sensory stimuli, addressing distractions, and driving attention towards the experience's intention. This will contribute to designing VR experiences such as learning, task performance, and behavior change. This study also attempts to extend the knowledge of human sensory perception in simulated human environments.

6 METHODS AND TOOLS

Regarding the measurements for presence, we will adopt SUS (Slater, Usoh and Steed) [3] questionnaire inside the virtual environment [14] due to its validity discussed in the literature. For attention, we are conducting selective attention [15] exercise

applying a retention and transfer test at the end of the task. At the same time, corroborative physiological measurements such as EEG [16] will be applied for attention, focus, and memory.

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